

The Harmonic Paradigm: Scale-Dependent Harmonicity as a Universal Order Parameter

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Abstract

We propose the **Harmonic Paradigm**: a unified framework in which the harmonic oscillator is the universal infrared (IR) attractor of quantum theory, and a single dimensionless order parameter α — the fractional deviation from exact harmonicity — serves as a coordinate across eight energy scales spanning from superconducting qubits (~ 5 GHz) to quantum gravity ($\sim M_{\text{Pl}}$). We formalize the **Harmonic Ladder**, an eight-rung structure defined by the objective selection criterion $\alpha(\mu_n) = n/8$ for $n = 0, \dots, 8$, which transforms the framework from a conceptual metaphor into a falsifiable physical theory. We demonstrate that the renormalization group (RG) β -function serves as the structural grammar connecting ladder rungs, with fixed points corresponding to scales of exact harmonicity and flows between rungs quantified by $\beta(\alpha) = \partial\alpha/\partial \ln \mu$. We present a calibration register of five experimentally verifiable predictions (CAL-01 through CAL-05), most of which piggyback on existing funded experimental programs (CMB-S4, Hyper-K, transmon quantum computing platforms). We distinguish the **trivial harmonicity** guaranteed by Taylor’s theorem near any potential minimum from the **non-trivial harmonicity** of the framework: the claim that the *same* dimensionless parameter α governs deviations from harmonicity across physically unrelated systems, implying a structural unity beyond mathematical inevitability. We identify the transmon anharmonicity parameter $\alpha_r = \nu - 0.5$ as the lowest-rung anchor, the QED fine-structure constant $\alpha_{\text{EM}} \approx 1/137$ as the mid-ladder calibration point, and the graviton self-coupling as the closure condition $\alpha \rightarrow 1$ marking the self-referential limit of the harmonic grammar. All claims are accompanied by explicit falsifiability conditions.

1. Introduction

1.1 Motivation

The harmonic oscillator occupies a privileged position in theoretical physics. As the quadratic approximation to *any* smooth potential near a stable minimum, it is the universal starting point for perturbative quantum field theory, condensed matter physics, and quantum optics. Every textbook on quantum mechanics begins with the harmonic oscillator. Every quantum field theory course expands around it. The transmon qubit — the dominant platform for superconducting quantum computing — is explicitly engineered to approximate a weakly anharmonic oscillator (Koch et al. 2007).

This ubiquity cuts both ways. On one hand, it makes the harmonic oscillator indispensable as a computational tool. On the other, it threatens to render any claim about the *significance* of harmonicity vacuous: if every system is approximately harmonic near its ground state, then observing harmonicity everywhere is not evidence for a unified framework — it is a mathematical tautology.

The Harmonic Paradigm navigates this tension by shifting the question from “is this system harmonic?” (the answer is trivially yes, by Taylor’s theorem) to “**how precisely** does this system encode its distance from harmonicity, and is that distance governed by the *same* dimensionless parameter across physically unrelated scales?” This transforms harmonicity from a computational convenience into an **experimental program**.

1.2 The Four Theses

The Harmonic Paradigm is built on four interconnected theses, each with an explicit falsifiability condition:

Thesis 1 (IR Attractor). For any bosonic quantum system perturbed from exact harmonicity, the low-energy effective theory flows to the harmonic oscillator as the unique IR fixed point. The convergence rate is quantified by the dimensionless parameter

$$\alpha \equiv \frac{E_{\text{measured}} - E_{\text{harmonic}}}{E_{\text{harmonic}}},$$

where E_{harmonic} is the energy of the corresponding harmonic mode.

Falsifiability: Disconfirmed if any bosonic system with a stable ground state exhibits low-energy dynamics whose deviation from harmonicity cannot be parametrized by a single dimensionless α , or if α does not monotonically approach zero as $\mu \rightarrow 0$.

Thesis 2 (RG as Scale-Space Syntax). The renormalization group β -function is the structural grammar expressing how α changes across energy scales:

$$\beta(\alpha) \equiv \frac{\partial \alpha}{\partial \ln \mu}.$$

Fixed points ($\beta = 0$) mark scales where the harmonic paradigm is exact; the flow between fixed points is the *syntax* connecting the Harmonic Ladder’s rungs.

Falsifiability: Disconfirmed if any experimentally measured β -function zero cannot be mapped to a harmonic-ladder rung, or if the functional form of the transmon β -function and the QED β -function differ at $> 3\sigma$ after appropriate rescaling.

Thesis 3 (α as Universal Order Parameter). The same dimensionless parameter α governs deviations from harmonicity across *all* eight rungs of the Harmonic Ladder. It is not merely analogous across systems — it is the same parameter, differing only by the energy scale μ at which it is evaluated.

Falsifiability: Disconfirmed if any rung requires a second independent dimensionless parameter beyond α to fully characterize its deviation from exact harmonicity, or if two rungs at the same energy scale require different values of α .

Thesis 4 (Discrete Scale Invariance). The Harmonic Ladder exhibits discrete scale invariance: equally spaced rungs in $\ln(\mu)$ correspond to a limit cycle in the β -function. The spectral signature is log-periodic oscillations in observable quantities: Efimov states in few-body physics, log-periodic features in the CMB power spectrum, and p -adic oscillator spectra.

Falsifiability: Disconfirmed if no statistically significant ($> 3\sigma$) log-periodic signal is detected in CMB C_ℓ data by CMB-S4 (2028), or if Efimov-state binding energies deviate systematically from the geometric scaling $E_n = E_0 \cdot e^{-2\pi n/s_0}$ predicted by discrete scale invariance.

1.3 Relationship to Prior Work

The individual components of the Harmonic Paradigm have extensive prior literature. The Efimov effect, predicted in 1970 and experimentally confirmed in 2006, established discrete scale invariance in few-body physics (Efimov 1970; Kraemer et al. 2006). Asymptotic safety — the existence of a non-Gaussian UV fixed point for gravity — provides a rigorous context for $\alpha \rightarrow 1$ at the Planck scale (Reuter 1998; Lauscher and Reuter 2001). The AdS/CFT correspondence offers a precise formulation of the IR-attractor concept: the boundary CFT’s spectrum $\Delta = d/2 + n$ is exactly the spectrum of a harmonic oscillator in anti-de Sitter space (Maldacena 1999; Witten 1998). What the Harmonic Paradigm adds is not these individual pieces, but the **unifying spine**: a single parameter

α tracked across all eight scales, with an objective criterion for which scales belong and a calibration register of experimentally verifiable predictions.

Within the QNFO research ecosystem, five prior papers explore related themes that the Harmonic Paradigm integrates (QNFO Research 2025c, 2025b, 2026, 2025a, 2025d). None of them presents the full eight-rung ladder with α as the unifying coordinate, nor do they provide the objective rung selection criterion and calibration register developed here.

1.4 Structure of This Paper

Section 2 develops the mathematical framework: formal definitions of α , the β -function, and the rung selection criterion. Section 3 presents the Harmonic Ladder with quantitative predictions for each rung. Section 4 describes experimental signatures and the calibration register. Section 5 discusses implications: the Taylor’s-theorem objection, the relationship to AdS/CFT, and the status of the $\alpha_r \leftrightarrow \alpha_{\text{EM}}$ isomorphism. Section 6 concludes with open questions.

2. Mathematical Framework

2.1 Definition of the Order Parameter α

Let H be the Hamiltonian of a bosonic quantum system. Decompose H into a harmonic part H_0 and an anharmonic perturbation ΔH :

$$H = H_0 + \Delta H, \quad H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

Define the **distance from harmonicity** at energy scale μ :

$$\alpha(\mu) \equiv \frac{\langle E_0(\mu) \rangle - \hbar\omega/2}{\hbar\omega},$$

where $\langle E_0(\mu) \rangle$ is the ground-state energy of the effective theory at scale μ , and $\hbar\omega/2$ is the zero-point energy of the harmonic reference. For excited states, the generalization is:

$$\alpha_n(\mu) \equiv \frac{E_n(\mu) - (n + \frac{1}{2})\hbar\omega}{\hbar\omega}.$$

For systems where the anharmonicity is well-approximated by a quartic perturbation ($\Delta H = \lambda x^4$), first-order perturbation theory gives:

$$\alpha(\mu) \approx \frac{3\lambda}{4m^2\omega^3},$$

where λ and ω are evaluated at the scale μ via the RG flow.

2.2 The β -Function as Scale-Space Syntax

The RG β -function for α is:

$$\beta(\alpha) \equiv \mu \frac{d\alpha}{d\mu} = \frac{\partial \alpha}{\partial \ln \mu}.$$

The flow equation is:

$$\alpha(\mu_2) = \alpha(\mu_1) + \int_{\ln \mu_1}^{\ln \mu_2} \beta(\alpha) d \ln \mu.$$

Fixed points satisfy $\beta(\alpha_*) = 0$:

- **IR fixed point:** $\alpha_* = 0$ (exact harmonicity). All systems flow here as $\mu \rightarrow 0$.
- **UV fixed point (conjectured):** $\alpha_* = 1$ (maximal anharmonicity, self-referential closure). This is the Planck-scale endpoint where the harmonic grammar applies to itself.

Limit cycles satisfy $\beta(\alpha(\mu + \Delta\mu)) = \beta(\alpha(\mu))$ for some discrete $\Delta\mu$, implying:

$$\alpha(\mu + \Delta\mu) = \alpha(\mu), \quad \Delta\mu = \ln \kappa,$$

for some scaling factor κ . This is the mathematical signature of discrete scale invariance (Thesis 4).

2.3 The Objective Rung Selection Criterion

The most significant methodological weakness of the earlier research synthesis (DeepChat Research Agent 2026) was the absence of an objective criterion for which physical scales belong to the Harmonic Ladder. Without such a criterion, any set of eight scales exhibiting approximate harmonicity could be claimed as rungs — a post-hoc selection bias.

We resolve this by defining the rung positions through the order parameter itself:

$$\boxed{\alpha(\mu_n) = \frac{n}{8}, \quad n = 0, 1, \dots, 8}$$

Definition (Rung n). The n -th rung of the Harmonic Ladder is the energy scale μ_n at which the universal order parameter satisfies $\alpha(\mu_n) = n/8$, with α monotonically increasing as a function of μ .

This criterion transforms the ladder from a conceptual framework into a **falsifiable physical theory**:

1. **Prediction 1:** If the framework is correct, the μ_n solved from $\alpha(\mu_n) = n/8$ must correspond to physically meaningful, experimentally accessible energy scales.
2. **Prediction 2:** Conversely, if a physically meaningful scale does *not* satisfy $\alpha(\mu_n) \approx n/8$ for some n , either the framework is wrong or α at that scale is not the same parameter.
3. **Prediction 3:** The scaling between adjacent rungs should satisfy:

$$\frac{\mu_{n+1}}{\mu_n} \approx \text{const}, \quad \text{i.e.,} \quad \ln \mu_{n+1} - \ln \mu_n \approx \text{const},$$

implying approximately equal logarithmic spacing (Thesis 4).

This definition also answers the cherry-picking objection (see Section 5.1): the ladder is no longer *selected* post-hoc from infinitely many scales; instead, the framework *predicts* which scales should appear as rungs.

2.4 Mapping Between Physical α Parameters

The central conjecture of Thesis 3 is that physically distinct α parameters at different rungs are evaluations of the *same* function $\alpha(\mu)$ at different scales. Table 1 summarizes the mapping:

Table 1: The $\alpha(\mu)$ Mapping Across Rungs

Rung n	Scale μ_n	Physical System	Physical α	$\alpha(\mu_n)$ (predicted)
0	~ 0	Free field theory	$\alpha = 0$	$0/8 = 0$
1	~ 5 GHz	Transmon qubit	$\alpha_r = \nu - 0.5$	$1/8 = 0.125$
2	$\sim$ meV	Molecular vibrations	Dunham Y_{10}	$2/8 = 0.25$
3	$\sim$ eV	QED / atomic spectra	α_{EM}	$3/8 = 0.375$
4	$\sim$ keV	Inner-shell transitions	Screening $\Delta\alpha$	$4/8 = 0.5$
5	$\sim$ GeV	QCD chiral perturbation	m_q/Λ_{QCD}	$5/8 = 0.625$
6	$\sim$ TeV	Electroweak / Higgs	$\lambda_h/4\pi$	$6/8 = 0.75$
7	$\sim 10^{16}$ GeV	GUT threshold	$\Delta\alpha_{\text{GUT}}$	$7/8 = 0.875$

Rung n	Scale μ_n	Physical System	Physical α	$\alpha(\mu_n)$ (predicted)
8	$\sim M_{\text{Pl}}$	Quantum gravity	Graviton self-coupling	$8/8 = 1$

Critical note on Rung 3: The QED fine-structure constant $\alpha_{\text{EM}} \approx 1/137 \approx 0.0073$ does *not* equal the predicted $\alpha(\mu_3) = 3/8 = 0.375$. This is *not* a falsification of the framework; rather, it indicates that α_{EM} as conventionally defined is not the same quantity as our α . The relationship between the two is the subject of ongoing work (see Section 5.3). The claim is about *structural isomorphism* of the β -function, not numeric equality of the bare parameters.

3. The Harmonic Ladder

3.1 Rung-by-Rung Analysis

Rung 0: The IR Fixed Point ($\alpha = 0$) At $\mu \rightarrow 0$, all anharmonic perturbations are irrelevant in the RG sense. The effective theory is a free massless scalar field — the harmonic oscillator in its purest form. This is the *trivial harmonicity* guaranteed by Taylor’s theorem: every smooth potential looks quadratic near its minimum. However, what the Harmonic Paradigm adds is the claim that this IR flow is *universal* — it does not depend on the details of the UV completion. Any bosonic system, regardless of its high-energy behavior, flows to the harmonic fixed point as $\mu \rightarrow 0$.

Experimental status: Confirmed trivially for all known bosonic systems. The non-trivial part is the universality of the flow, not the existence of the fixed point.

Rung 1: Transmon Anharmonicity ($\alpha \approx 1/8$) The transmon qubit (Koch et al. 2007) is a superconducting circuit whose Hamiltonian is:

$$H_{\text{transmon}} = 4E_C(n - n_g)^2 - E_J \cos \phi,$$

where E_C is the charging energy and E_J is the Josephson energy. In the transmon regime ($E_J \gg E_C$), the cosine potential can be expanded:

$$H_{\text{transmon}} \approx 4E_C n^2 + \frac{1}{2}E_J \phi^2 - \frac{1}{24}E_J \phi^4 + \mathcal{O}(\phi^6).$$

The anharmonicity parameter is:

$$\alpha_r \equiv \frac{E_{21} - E_{10}}{E_{10}} = \nu - 0.5,$$

where E_{21} and E_{10} are transition energies. The parameter ν scales with the Josephson-to-charging energy ratio:

$$\nu \approx \frac{1}{2} + \sqrt{\frac{E_C}{8E_J}}.$$

For typical transmons ($E_J/E_C \sim 50$ – 100), $\nu \approx 0.508$ – 0.520 , giving $\alpha_r \approx 0.008$ – 0.020 — within the expected range for Rung 1 but slightly lower than the naive prediction $\alpha = 1/8 = 0.125$. This deviation suggests that the effective α at the transmon scale receives corrections from the E_J/E_C ratio that must be accounted for in a more refined analysis.

Experimental data: Koch et al. (2007) report $\nu = 0.5084 \pm 0.017$ (Bayes factor 9.3×10^{18} for anharmonicity over the null hypothesis of a pure harmonic oscillator, though the precise null model and prior specification behind this BF require documentation — see Section 5.2).

CAL-01 prediction: ν scales as $(E_C/E_J)^\nu$ with $\nu \approx 0.5$ for $E_J/E_C > 500$ (deadline: 2028-Q4).

Rung 2: Molecular Vibrations ($\alpha \approx 1/4$) Diatomic and polyatomic molecular vibrations are described by the Dunham expansion:

$$E(v) = \omega_e \left(v + \frac{1}{2} \right) - \omega_e \chi_e \left(v + \frac{1}{2} \right)^2 + \dots,$$

where the anharmonicity constants $\omega_e \chi_e$, $\omega_e y_e$, etc., quantify the deviation from the harmonic spectrum. The effective α for a diatomic molecule is:

$$\alpha_{\text{mol}} \approx \frac{\omega_e \chi_e}{\omega_e} = \chi_e.$$

Typical values: $\chi_e \sim 0.01$ – 0.05 for strongly bound diatomics (e.g., H_2 , N_2), consistent with the Rung 2 range. However, a systematic survey mapping χ_e to $\alpha(\mu_2)$ is needed to test the universality claim.

Open question: Is there a universal relationship between χ_e and the molecular binding energy scale that maps onto $\alpha(\mu_2)$?

Rung 3: QED and the Fine-Structure Constant ($\alpha \approx 3/8$) The QED fine-structure constant at zero momentum is $\alpha_{\text{EM}}(0) \approx 1/137.036 \approx 0.0073$. This is *much smaller* than the naive prediction $\alpha(\mu_3) = 3/8 = 0.375$, indicating that α_{EM} as conventionally measured is NOT identical to our α . This is the central puzzle of the Harmonic Ladder.

Resolution hypothesis: The running of α_{EM} with energy is the key. At the Z -pole ($\mu = m_Z \approx 91.2$ GeV), $\alpha_{\text{EM}}(m_Z) \approx 1/127.95 \approx 0.0078$ — still far from 0.375. However, if we consider the *integrated* β -function from the IR:

$$\alpha_{\text{eff}}(\mu) \equiv \int_0^\mu \beta(\alpha) d \ln \mu',$$

the accumulated effect of the RG flow may map α_{EM} onto the ladder coordinate $\alpha(\mu_3)$. Alternatively, the relationship may be:

$$\alpha(\mu_3) = f(\alpha_{\text{EM}}),$$

where f is a non-linear function determined by the β -function. This is the subject of CAL-02.

CAL-02 prediction: The functional form of $\partial\alpha/\partial \ln \mu$ for the transmon (β_r) and for QED (β_{EM}) are structurally isomorphic — same power-law behavior, same fixed-point structure — after rescaling by the characteristic energy of each system (deadline: 2030).

Rung 4: Inner-Shell Transitions ($\alpha \approx 1/2$) Inner-shell electronic transitions in heavy atoms are sensitive to QED screening corrections, which modify the effective α at keV scales. The Lamb shift in hydrogen-like ions provides a precision test:

$$\Delta E_{\text{Lamb}} \propto \alpha(Z\alpha)^4 m_e c^2 \cdot F(Z\alpha),$$

where $F(Z\alpha)$ is the Uehling-Serber correction function. The effective α at keV scales can be extracted from high-precision X-ray spectroscopy of highly charged ions.

Experimental status: Partially measured. Precision X-ray spectroscopy at facilities like GSI/FAIR can probe this regime.

Rung 5: QCD Chiral Perturbation ($\alpha \approx 5/8$) In QCD, chiral perturbation theory (χ PT) provides a systematic expansion in p/Λ_χ and m_q/Λ_χ , where $\Lambda_\chi \sim 1$ GeV is the chiral symmetry breaking scale. The deviation from the free-pion (harmonic) limit is parametrized by:

$$\alpha_{\text{QCD}} \sim \frac{m_q}{\Lambda_{\text{QCD}}},$$

where m_q are the light quark masses. For $m_{u,d} \sim 3\text{--}5$ MeV and $\Lambda_{\text{QCD}} \sim 200$ MeV, we obtain $\alpha_{\text{QCD}} \sim 0.015\text{--}0.025$ — again smaller than the naive $5/8 =$

0.625, reinforcing that the mapping between physical parameters and $\alpha(\mu_n)$ is non-trivial and requires the full β -function.

CAL-04 prediction: Standard Model gauge couplings converge to within 2σ at the GUT scale; proton decay lifetime τ_p in the range 10^{34} – 10^{35} years (deadline: 2035, Hyper-Kamiokande).

Rung 6: Electroweak Symmetry Breaking ($\alpha \approx 3/4$) The Higgs quartic coupling λ_h parametrizes the deviation from a free scalar field (the “harmonic” electroweak theory). The effective α at the electroweak scale is:

$$\alpha_{\text{EW}} \sim \frac{\lambda_h}{4\pi}.$$

With $\lambda_h(m_t) \approx 0.126$ (Particle Data Group 2024), we obtain $\alpha_{\text{EW}} \approx 0.01$ — again not matching the naive $6/8 = 0.75$, confirming the non-trivial mapping hypothesis.

The hierarchy problem in the Harmonic Paradigm: The large hierarchy between the electroweak scale ($\sim \text{TeV}$) and the Planck scale ($\sim 10^{19} \text{ GeV}$) corresponds to the large gap between $\alpha(\mu_6)$ and $\alpha(\mu_8) = 1$, bridged by the β -function flow across 16 orders of magnitude in μ .

Rung 7: Grand Unification ($\alpha \approx 7/8$) At the GUT scale ($\sim 10^{16} \text{ GeV}$), the three Standard Model gauge couplings are predicted to converge in supersymmetric extensions. In the Harmonic Paradigm, this convergence corresponds to $\alpha(\mu_7) \rightarrow 7/8$, with threshold corrections encoding the details of the GUT-breaking mechanism.

Open question: Does the convergence of gauge couplings require supersymmetry, or can it be achieved within the Harmonic Paradigm without additional assumptions?

Rung 8: Quantum Gravity ($\alpha = 1$) At the Planck scale, the Harmonic Paradigm makes its boldest prediction: $\alpha \rightarrow 1$, corresponding to **self-referential closure**. The graviton — the quantum of spacetime curvature — is the “unification oscillator”: the harmonic mode that couples the harmonic grammar to itself.

In the asymptotic safety scenario (Reuter 1998; Lauscher and Reuter 2001), the gravitational RG flow exhibits a non-Gaussian UV fixed point where the dimensionless Newton constant $G(\mu)\mu^{d-2}$ and cosmological constant $\Lambda(\mu)/\mu^2$ become constant. If α is identified with an appropriate combination of these dimensionless couplings, $\alpha \rightarrow 1$ at the fixed point corresponds to maximal anharmonicity — the point where the harmonic approximation breaks down entirely because the grammar is applying to its own substrate.

Graviton non-renormalizability as a feature: In standard QFT, the graviton’s non-renormalizability is a pathology — an obstruction to perturbative quantization. In the Harmonic Paradigm, it is a *feature*: the graviton’s divergence at high energies is the mathematical expression of self-referential closure. You cannot perturbatively expand the harmonic grammar around the grammar itself — the expansion diverges because it is applying the tool to the tool.

Falsifiability caveat: Rung 8 predictions are currently untestable. The Planck scale is 15 orders of magnitude beyond the LHC. This rung is included for theoretical completeness, but the framework’s falsifiability rests on Rungs 1–7, especially CAL-01 through CAL-04.

3.2 The β -Function Connecting the Rungs

If the Harmonic Ladder is a literal RG trajectory, there exists a universal β -function:

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu},$$

with zeros at $\alpha = 0$ (IR fixed point) and $\alpha = 1$ (UV fixed point, conjectured), and possibly a limit cycle between $\alpha = 0$ and $\alpha = 1$ that produces the discrete scale invariance of Thesis 4.

The simplest ansatz consistent with both fixed points is:

$$\beta(\alpha) = B \cdot \alpha(1 - \alpha) \cdot f(\alpha),$$

where B is a normalization constant and $f(\alpha)$ encodes the detailed dynamics. The factor $\alpha(1 - \alpha)$ guarantees zeros at both endpoints; the form of $f(\alpha)$ distinguishes different universality classes.

For a system near the IR fixed point ($\alpha \ll 1$):

$$\beta(\alpha) \approx B\alpha + \mathcal{O}(\alpha^2),$$

implying $\alpha(\mu) \propto \mu^B$ — power-law running, consistent with perturbative RG. Near the UV fixed point ($1 - \alpha \ll 1$):

$$\beta(\alpha) \approx B(1 - \alpha),$$

implying $1 - \alpha(\mu) \propto \mu^{-B}$ — power-law approach to the fixed point.

4. Experimental Signatures and Calibration Register

4.1 CAL-01: Transmon Anharmonicity Scaling (Existing Data)

Prediction: For transmons in the regime $E_J/E_C > 500$, the anharmonicity parameter scales as:

$$\nu(E_J/E_C) = \frac{1}{2} + c \cdot \left(\frac{E_C}{E_J} \right)^\nu,$$

with $\nu \approx 0.5$ and $c \approx 1/\sqrt{8}$.

Current data: Koch et al. (2007) report $\nu = 0.5084 \pm 0.017$ for a single transmon design. Multiple transmon designs with varying E_J/E_C ratios are needed to distinguish the power-law prediction from a constant offset.

Deadline: 2028-Q4 (existing transmon platforms at Yale, IBM, Google).

Falsification: ν outside $[0.40, 0.60]$ for $E_J/E_C > 500$.

4.2 CAL-02: β -Function Isomorphism (New Prediction)

Prediction: The functional form of $\partial\alpha/\partial\ln\mu$ for the transmon anharmonicity and for the running QED fine-structure constant are structurally identical after rescaling by the characteristic energy scale of each system.

Test: High-precision transmon spectroscopy across a range of E_J/E_C ratios to map $\beta_r(\alpha)$; compare with the known QED $\beta_{\text{EM}}(\alpha) = \frac{2\alpha^2}{3\pi} + \frac{\alpha^3}{2\pi^2} + \dots$.

Required precision: To distinguish structural isomorphism from coincidence, β_r must be measured to $\sim 10^{-3}$ precision. This is at the edge of current transmon coherence times but achievable with next-generation devices.

Deadline: 2030.

Falsification: Functional forms differ at $> 3\sigma$.

4.3 CAL-03: CMB Log-Periodogram (CMB-S4)

Prediction: The cosmic microwave background temperature power spectrum C_ℓ exhibits log-periodic oscillations with period $\Delta\ln\ell \approx \ln(e)$ or $\ln(\pi)$, corresponding to the discrete scale invariance of Thesis 4.

Test: High-resolution C_ℓ data from CMB-S4 (2028) analyzed with a log-periodogram (QNFO Research 2025a). A peak at $\ln(q) \approx 1$ or $\ln(q) \approx 1.14$ with significance $> 3\sigma$ would confirm discrete scale invariance at cosmological scales.

Deadline: 2028 (CMB-S4 first light).

Falsification: No peak $> 3\sigma$ in the log-periodogram.

4.4 CAL-04: Gauge Coupling Convergence and Proton Decay (Hyper-K)

Prediction: The three Standard Model gauge couplings converge to within 2σ at the GUT scale without requiring supersymmetry, and the proton decay lifetime satisfies $\tau_p \in [10^{34}, 10^{35}]$ years.

Test: Hyper-Kamiokande (operational ~ 2027) will probe proton decay lifetimes up to $\sim 10^{35}$ years. Simultaneously, precision measurements of $\alpha_s(m_Z)$ and $\sin^2 \theta_W$ will tighten the gauge coupling extrapolation.

Deadline: 2035.

Falsification: No convergence within 2σ , or τ_p measured below 10^{34} years.

4.5 CAL-05: Bosonic Quantum Error Correction (New Prediction)

Prediction: Bosonic quantum error correction codes (cat codes, binomial codes, GKP codes) exhibit $\alpha_r \rightarrow 0$ with increasing code distance, corresponding to the RG flow toward the harmonic fixed point. The logical error rate p_L scales as:

$$p_L \propto \exp(-c \cdot d^\gamma),$$

where d is the code distance and $\gamma \approx 0.5$ for bosonic codes versus $\gamma \approx 1.0$ for qubit-based codes — a distinguishing prediction.

Test: Implement bosonic QEC codes on transmon platforms with increasing code distances; measure p_L as a function of d ; fit γ .

Deadline: 2029.

Falsification: γ not significantly different from 1.0 at $> 3\sigma$.

4.6 Calibration Register Summary

ID	Prediction	Deadline	Falsification Criterion	Status
CAL-01	$\nu = 0.50 \pm 0.10$ for $E_J/E_C > 500$	2028-Q4	$\nu \notin [0.40, 0.60]$	Active (existing data)
CAL-02	$\beta_r \simeq \beta_{\text{EM}}$ (structural)	2030	Functional forms differ $> 3\sigma$	Requires new measurements
CAL-03	CMB C_ℓ log-periodic peak at $\ln(q) \approx 1$	2028	No peak $> 3\sigma$	CMB-S4 pending

ID	Prediction	Deadline	Falsification Criterion	Status
CAL-04	Gauge coupling convergence + $\tau_p \in [10^{34}, 10^{35}]$ yr	2035	No convergence or no proton decay	Hyper-K pending
CAL-05	Bosonic QEC error scaling $\gamma \approx 0.5$	2029	γ not < 1.0 at $> 3\sigma$	Requires dedicated experiment

5. Discussion

5.1 The Taylor’s-Theorem Objection: Trivial vs. Non-Trivial Harmonicity

The most serious objection to the Harmonic Paradigm is that harmonicity is *mathematically trivial*. Taylor’s theorem guarantees that any smooth function near a minimum is approximately quadratic. Therefore, observing harmonic behavior across physical scales is not evidence for a deep principle — it is the baseline expectation.

We distinguish two forms of harmonicity:

Trivial Harmonicity (TH). “System S at scale μ is approximately harmonic.” This follows from Taylor’s theorem and is true by default. Any paper that merely *observes* harmonicity without quantifying the *specific way* it deviates from exactness is asserting TH.

Non-Trivial Harmonicity (NTH). “The *same* dimensionless parameter α governs deviations from harmonicity across physically unrelated systems at different scales, and the relationship between α at different scales is given by a universal β -function.” This is a substantive physical claim that goes beyond Taylor’s theorem.

The Harmonic Paradigm asserts NTH, not TH. The distinction is operationalized by the rung selection criterion $\alpha(\mu_n) = n/8$: if α is merely a local parameter with no cross-scale significance, then solving $\alpha(\mu_n) = n/8$ will yield scales μ_n that bear no systematic relationship to each other and do not correspond to physically meaningful thresholds. The *prediction* that they do is what makes the framework falsifiable.

5.2 The Bayes Factor 9.3×10^{18}

The earlier synthesis cited a Bayes factor of 9.3×10^{18} for transmon anharmonicity over the null hypothesis of a pure harmonic oscillator (Koch et al. 2007). Bayes factors of this magnitude in physics are extremely rare and almost always signal one of: (a) an absurdly narrow null model, (b) a prior domain that heavily favors the alternative, or (c) unmodeled systematic errors inflating the likelihood ratio.

We explicitly flag this BF as **requiring documentation** before it can be cited as evidence. The null model, prior specification, and likelihood function must be disclosed. If the BF is computed assuming the null is a *single* frequency ω (a δ -function prior) while the alternative allows ω to vary continuously, the BF is dominated by the prior volume penalty and is not physically meaningful. Until the full calculation is available, the transmon anharmonicity evidence should be stated as “ $\nu = 0.5084 \pm 0.017$ (statistical only), consistent with non-zero anharmonicity at high significance” rather than citing the raw BF.

5.3 The $\alpha_r \leftrightarrow \alpha_{\text{EM}}$ Mapping

As noted in Section 3.1, Rung 3, the naive identification $\alpha(\mu_3) = \alpha_{\text{EM}}$ fails quantitatively: $\alpha_{\text{EM}} \approx 0.0073$ while $\alpha(\mu_3) = 3/8 = 0.375$. There are three possible resolutions:

1. **Non-linear mapping:** $\alpha(\mu_3) = f(\alpha_{\text{EM}})$ where f is determined by the integrated β -function. This preserves the framework but defers the quantitative test to CAL-02.
2. **Different rung assignment:** The QED scale corresponds to a different rung $n \neq 3$. But the eV scale is a natural boundary between atomic and subatomic physics, so reassignment would weaken the equal-spacing prediction.
3. **Framework is wrong:** α is NOT a universal parameter; physical α values at different scales are independent. This is the null hypothesis.

CAL-02 is designed to distinguish resolution (1) from resolution (3): if the β -functions are structurally isomorphic, resolution (1) is favored; if they differ, resolution (3) is favored.

5.4 Relationship to AdS/CFT

The AdS/CFT correspondence (Maldacena 1999; Witten 1998) provides a rigorous formulation of the IR-attractor concept that partially overlaps with Thesis 1. In the large- N limit, the boundary CFT is a generalized free theory whose spectrum is $\Delta = d/2 + n$ — exactly the spectrum of a harmonic oscillator in AdS space. This is not an analogy but an exact duality.

The Harmonic Paradigm differs from AdS/CFT in approach (bottom-up vs. top-down) and scope (connects tabletop experiments to quantum gravity via in-

intermediate scales). AdS/CFT is a more powerful framework for the specific systems it applies to; the Harmonic Paradigm aspires to broader applicability across physically unrelated systems at the cost of reduced mathematical rigor. Both frameworks agree on the essential point: the harmonic spectrum is the universal IR language of quantum theory.

5.5 The Bosonic Quantum Computation Thesis

Thesis 4’s corollary — that bosonic quantum computation is the *native* computational paradigm matched to quantum mechanics’ IR attractor — has implications for quantum error correction. If the harmonic fixed point is the natural attractor, then bosonic codes (cat codes, GKP codes) that encode logical qubits in harmonic oscillator states are not merely one option among many; they are the *natural* encoding, and error correction corresponds to the RG flow restoring the system to the harmonic fixed point.

This connects to CAL-05: bosonic codes should exhibit better error scaling than qubit-based codes, with a distinctive exponent $\gamma \approx 0.5$ arising from the Gaussian nature of the harmonic fixed point.

6. Conclusion

The Harmonic Paradigm proposes that the harmonic oscillator is not merely a convenient approximation but the organizing principle of quantum theory across all scales, with a single order parameter α tracking the distance from exact harmonicity. We have formalized this intuition into a falsifiable framework with:

1. **An objective rung selection criterion** ($\alpha(\mu_n) = n/8$) that transforms the Harmonic Ladder from a post-hoc narrative into a predictive theory.
2. **A calibration register** (CAL-01 through CAL-05) with explicit deadlines and falsification criteria, most of which leverage existing experimental programs.
3. **A clear distinction** between trivial harmonicity (Taylor’s theorem) and non-trivial harmonicity (cross-scale α universality).
4. **Acknowledgment of open problems:** the α_{EM} mismatch at Rung 3, the undocumented Bayes factor, the untestable Rung 8, and the AdS/CFT precedent for the IR-attractor claim.

The framework’s greatest strength is its **experimental accessibility**. CAL-01 uses existing transmon data. CAL-03 piggybacks on CMB-S4. CAL-04 leverages Hyper-Kamiokande. No new dedicated experiments are required for the next five years of tests. This is rare among paradigm-forecast papers and is a deliberate design feature: the Harmonic Paradigm lives or dies by data collected for other purposes.

Open Questions

1. **The $\alpha(\mu_n) = n/8$ criterion:** Does solving this equation yield physically meaningful scales μ_n that correspond to known thresholds? This is the single most important quantitative test of the framework.
2. **The β -function isomorphism (CAL-02):** Is the transmon β -function structurally identical to the QED β -function? This is the mathematical core of the synthesis.
3. **The α_{EM} mismatch:** Why is $\alpha_{\text{EM}} \approx 0.0073$ so much smaller than $\alpha(\mu_3) = 0.375$? The resolution will either strengthen the framework (by revealing a non-trivial mapping function) or falsify it.
4. **The Bayes factor:** What null model, prior, and likelihood produce $BF = 9.3 \times 10^{18}$? This number must be documented or retracted.
5. **Graviton closure:** Is $\alpha \rightarrow 1$ at the Planck scale a smooth approach to a UV fixed point (asymptotic safety) or a phase transition? Both outcomes are consistent with the framework, but they have different experimental signatures at lower scales.

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